Nabanita Das

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Research interests

Studies of supermassive black holes, active galactic nuclei, galactic dynamics. Observations from Hubble, and ground-based NIR IFU spectroscopy.

Education

PhD in Astronomy — *August 2022 -*
Georgia State University — Atlanta, USA

Master's in Physics (Astronomy concentration) — *June 2024*
Georgia State University — Atlanta, USA

Master's in Physics — *July 2021*
Presidency University — Kolkata, India

Bachelor's in Physics — *July 2019*
Presidency University — Kolkata, India

Publications

3.1 Refereed

[4] **Das, N.**, Bentz M. C., Vasiliev E., Falcon-Barroso J., Valluri M., Onken, C., Raimundo, S., & Vestergaard, M., Beyond Gauss–Hermite: A Revised Stellar Dynamical Mass for MCG–06-30-15, in prep., expected submission date: December 2026.

[3] **Das, N.**, Bentz M. C., Vasiliev E., Valluri M., & Onken C., Revised Stellar Dynamical mass for the Central Black Hole in NGC 3227, submitted to ApJ, doi: 10.48550/arXiv.2611.05547

[2] **Das, N.**, Bentz M. C., Vasiliev E., Valluri M., Onken C., Raimundo, S. & Vestergaard M. (2025), A Stellar Dynamical Mass for the Central Black Hole in MCG–06-30-15. The Astrophysical Journal, Volume 993, Issue 1, id.11, doi: 10.3847/1538-4357/ae03a8

[1] Bhattacharyya J., Chatterjee R., Ghosh R. & **Das N.** (2020), Blazar Variability: A Study of Nonstationarity and the Flux–RMS Relation. The Astrophysical Journal, Volume 897, Issue 1, id.25, doi: 10.3847/1538-4357/ab91a8

3.2 Other

[1] Bentz, M., **Das N.**, +14 (2023), Broad-Band Photometric Monitoring of 2500 Alascattalo. *Minor Planet Bulletin*.

Invited Talks

[6] University of Surrey, November 20, 2026 (*virtual*).

[5] University of Michigan, November 11, 2026.

[4] Galaxy/AGN Journal club, Space Telescope Science Institute, October 28, 2026.

[3] Astronomy on Tap, Atlanta, April 20, 2026. Recorded talk.

[2] Lunch symposium at the Green Bank Observatory, February 23, 2025.

[1] Memorial University of Newfoundland astronomy club, January 30, 2025 (*virtual*).

Conferences

Contributed Talks

[7] Massive Black Holes across Cosmic Time, September 2025, Kavli Institute for Cosmology, Cambridge, UK.

[6] Young Astronomers for Galactic Nuclei Meeting, September 2025, University of Southampton, UK.

[5] 56th meeting of the Division on Dynamical Astronomy of American Astronomical Society, Atlanta, Georgia. Abstract here.

[4] Georgia Regional Astronomy Meeting, 2024

[3] Georgia Regional Astronomy Meeting, 2023

[2] 21st National Space Science Symposium of India, February 2022 (virtual).

[1] 3rd Undergraduate Symposium of Presidency University, *Best Talk*, July 2019, Kolkata, India, recorded talk.

Contributed Posters

[3] 243rd meeting of the American Astronomical Society. Abstract here.

[2] 40th meeting of Astronomical Society of India (virtual), February 2022.

[1] 39th meeting of Astronomical Society of India, February 2021.

Grants

2026 Collaborative Call for Proposals Award, Co-I, $10000 *2026-2027*

Georgia Space Grant Consortium

- NASA Space Grant Fellowship for astronomy related outreach programs in metro Atlanta and Georgia.

Successful Telescope Proposals

100m Green Bank Telescope, PI *46 hours*

Semester 2026A

100m Green Bank Telescope, PI *99 hours*

Semester 2025A

Observing Experience

100m Robert C. Byrd Green Bank Telescope *hours observed: 56*

on-site and remote observer, HI 21cm spectroscopy of a nearby disk galaxy

3.5m ARC Telescope at Apache Point Observatory *hours observed: 24*

on-site observing experience on KOSMOS spectrograph, long-slit spectroscopy of AGN

24" Miller Telescope at the Hard Labor Creek Observatory *hours observed: 21*

on-site observing experience, photometric studies for asteroids

Skills

Programming Languages: Python, IDL, Bash

Astronomical Tools: IRAF, GBTIDL, pPXF, Galfit, Matplotlib, FORSTAND, TOPCAT, Aladin

OS: Linux, MacOS, Windows

Honors and Awards

Graduate Travel Award *September 2025*

Georgia State University

- Travel grant of $ 500 awarded to present research at the Young astronomers on Galactic nuclei meeting.

3 Minute Thesis First Place Doctoral Award *March 2025*

Georgia State University

- Awarded $500 for winning first place in the 2025 3MT competition at Georgia State University.

Kanakprava Devi Memorial Award *Jan 2019*

Presidency University

- Awarded 2000 INR to the best resident undergraduate female student in natural sciences.

Leadership

Graduate Leader & Social Media Coordinator *July 2025 -*

GOT space, Astronomy on Tap, Atlanta

- Manage collaborations with partner organizations, create contents for social media accounts, and manage virtual interactions.
- Marketing of all endeavors of GOT Space (e.g., Cosmic Conversations at planetarium, Astronomy on Tap, ATL). On popular media: Astronomy on Tap Atlanta, Cosmic Conversations.
- Assist and train new volunteer ambassadors for outreach events.
- Volunteer for outreach events organized at K-12 schools, hosted 10+ events.

President of Women in Physics *August 2023 – July 2025*

Georgia State University

- Organize educational events for undergraduate women and minorities in physics at GSU, such as coffee hours with colloquia speakers and female faculty members in STEM.
- Arrange professional development panels to help navigate applications to graduate school and research internships.
- Coordinate and oversee social and recreational events for undergraduate students in Physics & Astronomy.

Astronomy Peer Advising Leaders: AstroPAL *August 2024 -*

Georgia State University

- Mentor 1st and 2nd year GSU astronomy students to navigate graduate school, and prepare for qualifying exams. Mentee: *Megan Peatt.*
- Create presentations for training new graduate students on several stuff (e.g., time management, budgeting, mentor-mentee relationships).

Astronomy Liaison *August 2025 - July 2026*

Physics Graduate Student Association, Georgia State University

- Assist other leaders in organization to coordinate events for graduate students in the department.
- Head of communication between astronomy graduate students and faculty members.

Teaching Experience

Teaching Assistant for ASTR1010 *Fall 2022, 2024, 2025 & Spring 2024-26*
Georgia State University

- Laboratory section for the introductory astronomy course on the solar system for non-major students.
- Typical class size: 24 students, taught 11 sections, *in-person & asynchronous.* **Independent and only instructor**

Teaching Assistant for ASTR1020 *Spring, Summer & Fall 2023*
Georgia State University

- Laboratory section for the introductory astronomy course on the astronomy outside the solar system.
- Typical class size: 24 students, taught 4 sections. **Independent and only instructor**

Guest lecturer for ASTR3520 *Spring 2026*
Georgia State University

- Taught one lecture on "Introduction to AGN" for upper level physics course on stars and galaxies.

Guest lecturer for PHYS1000 *Fall 2025*
Georgia State University

- Taught one lecture on "Research & Career Path" for the Gateway to Physics course.

Service

[7] Referee for *The Astrophysical Journal (ApJ)* since 2026.

[6] Graduate representative for Publicity & Outreach Committee, Department of Physics & Astronomy, Georgia State University, 2024-2025.

[5] Panelist in the professional development panel for REU students organized by Department of Physics & Astronomy, Georgia State University (2023-2024).

[4] Member of Physics Graduate Students' Association,GSU, 2023 - now.

[3] Graduate student representative for hiring committee of extragalactic astronomy, GSU, 2023-2024.

[2] Co-organizer and coordinator of webinar series "Alumni Lecture Series", Department of Physics, Presidency University (2020-2021).

[1] SOC and LOC of undergraduate Physics symposium at Presidency University, India (2017-2019).

References

Misty C. Bentz — Professor, Georgia State University
Ph.D. advisor | bentz@astro.gsu.edu

Monica Valluri — Research Professor, University of Michigan
Dissertation Committee member, Research Collaborator | mvalluri@umich.edu

Eugene Vasiliev — Reserch Fellow, University of Surrey
Research Collaborator | e.vasiliev@surrey.ac.uk